Using the AMD Ross™ Agentic AI Assistant for Vivado™ Design Suite

Course Agenda
2026.1

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Course Objectives

After completing this course, you will be able to:

- Explain how the AMD Ross™ agentic AI assistant integrates into programmable device design workflows
- Set up and validate a Ross-enabled AMD Vivado™ development environment
- Create and implement designs using Ross-guided workflows
- Analyze design data using linting, logs, and multi-run techniques
- Apply structured timing closure methodologies using Ross agentic AI assistance
- Debug simulation and protocol issues (e.g., AXI) using Ross-guided workflows
- Perform hardware debug using Ross-assisted ILA and VIO insertion

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Agenda

- AMD Ross™ Agentic AI Assistant for Vivado Design Suite Introduction {Lecture}
- Agentic AI Workflows for Advanced Adaptive SoC Design {Lecture}
- AMD Ross Agentic AI Assistant for Vivado Design Suite Knowledge Base Exploration {Lab}
- Spec-Driven Design Creation Prototyping on the Kria™ KV260 Vision AI Starter Kit {Lab}
- AMD Ross Agentic AI Assistant for Vivado Design Suite Design Analysis {Lab}
- AXI4 Protocol Debug and Ross-Driven Timing Closure {Lab}
- AXIS-ILA and AXIS-VIO Insertion Flow on the Versal™ VCK190 Evaluation Kit {Lab}

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Lab Environment

- Virtual machine running on a Windows® 10 or Windows 11 machine
  - Ubuntu® 22.04.5 LTS
- Vivado™ Design Suite
- Vivado AI Assistant Extension
- Visual Studio Code (VS Code)
- GitHub Copilot (for Ross™-assisted workflows)

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Prerequisites

- Basic knowledge of AMD FPGAs and adaptive SoCs
- Familiarity with Vivado™ Design Suite
- Understanding of RTL design and simulation concepts

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Latest Product Information

Please visit the following resources for the most current information on the AMD devices described in this course

- User design information can be found in user guides
- IP core information can be found in product guides
- Characteristics, such as timing, performance, etc., are listed in data sheets
- For design and software issues or bugs, see www.amd.com > Support
- DocNav provides easy access to these documents



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AMD Ross™ Agentic AI Assistant for Vivado™ Design Suite: Introduction

2026.1

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Agenda

1 **Agentic AI is Here**

- What the Ross™ agentic AI assistant for the Vivado™ tools is

2 **The AI Foundation**

- MCP + Agent Skills: The guiding principle

3 **Knowledge & Intelligence**

- Documentation search + knowledge base

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1 **Agentic AI is Here**

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What the Ross Agentic AI Assistant for the Vivado Tools Is

Tell the Vivado™ tool what to do — in plain English

What Can You Do?

1. **Search AMD Documentation — No Vivado Tool Needed**
2. **Drive the Vivado Tool with Natural Language**
3. **Run Skill-driven Workflows**

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1 Search AMD Documentation — No Vivado™ Tool Needed

System Architect

How to configure NoC for optimal bandwidth?

→ **UG1388, UG1504, PG313**

Use traffic classes (best effort, isochronous, low latency) with bandwidth allocation per route

RTL Developer

RTL coding best practices for synthesis?

→ **UG901, UG1387, UG949**

Follow UG901 guidelines. Use RAM_STYLE, SHREG_EXTRACT, MAX_FANOUT attributes. Avoid latches

Timing Closure Specialist

How to fix setup violations on high-fanout nets?

→ **UG949, UG1387, AR#53430**

Use report_high_fanout_nets after synthesis
Apply MAX_FANOUT or manual replication

Hardware Debug Specialist

PDI boot failure troubleshooting on AMD Versal™ devices?

→ **UG1304, UG908, AR#76655**

Check boot mode pins (SW1). Verify PDI in JTAG/flash/SD
Boot via HW Manager. Check PMC error codes

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2 Drive the Vivado™ Tool with Natural Language

"Open my project and show me the utilization summary"

Agent runs `open_project`, then `report_utilization`—presents LUT, FF, BRAM, DSP, and I/O usage as a clean table

Flags any resource above 80% utilization

“What are the worst 10 timing paths?”

Agent runs `report_timing_summary`, parses the output, and shows the critical paths ranked by slack—with source, destination, clock domain, and delay breakdown

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3 Run Skill-driven Workflows

"Run RTL lint and fix all issues“

Agent opens project, runs `synth_design -lint`, categorizes violations by severity, looks up fix guides in AMD docs, patches your RTL, and re-verifies—all in one prompt

"Analyze timing violations and generate fix strategies"

Agent follows a skill-driven methodology: classify failing paths by root cause, suggest constraint changes with real cell and net names, rerun implementation, and measure improvement—iteratively

"Compare all implementation runs"

Agent discovers four runs, extracts timing, congestion, and utilization from each, ranks them by QoR score, and generates an interactive dashboard with five visualization tabs

"Debug the AXI protocol violation"

Agent runs XSim, catches the assertion, opens a color-coded waveform, traces the root cause through the AXI handshake, and waits for your approval before fixing

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Three Ways Ross Accelerates FPGA Development

From Individual Productivity to Connected Intelligence

Accelerate Complex Design Workflows

Automate repetitive work to focus on differentiation

Scale Expertise Across the Team

Bring proven expertise into every developer's workflow

Connect Intelligence Across the Flow

Carry context from design through implementation and debug



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How Ross Supports Roles Across the FPGA Design Flow

Design Architect	Design Capture	Design Analysis	Design Closure	Hardware Debug
System Architect	RTL Developer	Verification Engineer	Timing Expert	Hardware Lab
CHALLENGE	**CHALLENGE**	**CHALLENGE**	**CHALLENGE**	**CHALLENGE**
• Slow design exploration & device selection	• Extensive Tcl & IPI hand-coding, no style enforcement	• Overwhelming logs & reports, manual triage across runs	• Iterative guesswork with low timing & DRC closure hit-rates	• Complex NoC, DDR, PCIe® link & PDI load debugging
PROMPT	**PROMPT**	**PROMPT**	**PROMPT**	**PROMPT**
• Explore Zynq™ UltraScale+™ options for my video pipeline use case • Compare AMD Versal™ vs Zynq for my AI inference workload	• Create IPI block design with Zynq PS, AXI GPIO & Timer • Check my RTL for best practices—style & pragmas	• 47 critical warnings—which ones matter and what to fix? • Identify critical paths & congestion hotspots post-synthesis	• 3 failing timing paths after 8 implementation runs—what next? • Suggest constraints for this clock domain crossing violation	• PDI boot fails on Versal VCK190 devices after adding NoC—what's wrong? • My DDR calibration fails at 3200 MT/s—debug steps?
OUTCOME	**OUTCOME**	**OUTCOME**	**OUTCOME**	**OUTCOME**
• RAG-backed guided exploration & device recommendations	• Natural language to IPI/RTL design + RTL best practices	• Prioritized issues with fixes, critical path & congestion	• Evidence-backed suggestions, improved closure hit-rate	• Rapid hardware bring up via agentic AI workflows

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- What the Ross™ agentic AI assistant for the Vivado™ tools is

2 **The AI Foundation**

- MCP + Agent Skills: The guiding principle

3 **Knowledge & Intelligence**

- Documentation search + knowledge base

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Key Components of Ross

Natural language interaction grounded in AMD expertise and connected tools

Client-Agnostic Interface

VS Code | Claude Code | Cursor | Devin | OpenCode CLI | Any MCP-Compatible Client

Ask → Interpret → Connect → Inform or Act

AMD Ross™ Agentic Workflow

Connection	**Model Context Protocol (MCP)**	Action	**Agent Skills**
AMD Vivado™ Design Suite | Vitis™ Tools | Docs		Autonomous Flows | Domain Expertise | Execution | Reports	
Context	**Vectorized Knowledge Base**	Flow	**Design Examples**

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Key Components of Ross: MCP & Agent Skills

Based on Open Standards

MCP gives the **information**

Agent skills give the **power to act**

Model Context Protocol

Open standard by Anthropic; adopted across the industry

What it provides:

- **Tool Discovery:** LLM discovers available AMD tools at runtime
- **Knowledge Access**: RAG-powered search across AMD docs
- **Session Control**: Start, stop & monitor AMD tool sessions
- **Execution Bridge:** Run any command via standardized API

Why it matters:

- **IDE-Agnostic:** VS Code, Cursor, Claude Code, Devin & any MCP client
- **LLM-Agnostic:** Claude, GPT, Gemini, Qwen & Llama — any model

Anthropic Open Standard

Markdown files encoding domain expertise as executable workflows

What it provides:

- **Autonomous Workflows:** Multi-step flows run end-to-end
- **Domain Expertise**: Hardware debug, timing closure & software optimization
- **Structured Execution:** Real commands, decision trees & no hallucination
- **Actionable Reports:** REPORT.md with real names & fix commands

Why it matters:

- **Repeatable:** Same skill, same results—eliminates tribal knowledge
- **Shareable:** Version-controlled, portable across teams & organizations

Works with :	VS Code	Cursor	Claude Code	Devin	OpenCode CLI	Any MCP-compatible client

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Framework Enabling Agentic AI Design Flow

MCP Clients — LLM-Driven IDEs & CLIs

- VS Code
- Cursor
- Claude Code
- Devin
- OpenCode CLI
- Any MCP-compatible client

Agent Skills execute here—Autonomous FPGA workflows powered by domain expertise

MCP Protocol

AMD Vivado™ IDE MCP Server — Local to Vivado Install

vivado_doc_search

RAG-powered search across AMD documentation

Example: "How to handle async clocks?"

- Vivado IDE
- Software Platform
- PDM
- ChipScope
- Model Composer

vivado_execute

Run any Tcl command through a standardized API

Example: "Analyze critical paths and recommend fixes"

- Tcl & GUI
- Project & Non-project
- Any Vivado IDE Version

AMD Knowledge Base — Local or Remote

- Multi-GB vectorized database with semantic search
- Tools including Vivado, Vitis, PDM, ChipScope, Model Composer
- Datasheets, white papers, answer records
- Tutorials, wiki, labs, and examples

Vivado Software — Local Installation

- Support for any previous version of Vivado Design Suite
- Full Tcl and GUI access
- Project & non-project mode
- Synthesis, implementation, bitstream

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Initial Delivery Scope: IDE/CLI Agnostic AMD Vivado MCP Server

Vivado™ MCP Server

vivado_doc_search
RAG-powered documentation search

vivado_execute
Run Tcl commands & GUI control

vivado_analyze
Reports, timing, CDC analysis

Vivado Tool Knowledge Base Local or Remote

Vivado & Vitis™ tools UG835, UG906, UG949	**PDM & ChipScope** Debug & program tools
Vitis Model Composer DSP & AI design docs	**Datasheets & More** Specs, wiki, ARs

- Multi-GB vectorized database with semantic search (RAG)
- Vivado IDE, Vitis software platform, PDM, ChipScope, Model Composer docs
- Runs as Docker® container—local or remote deployment
- Context-aware results ranked by relevance

MCP

MCP Client IDEs

Claude Desktop
Cursor
VS Code
Devin
Antigravity

MCP

MCP Client CLIs

Claude Code
Cursor CLI
GitHub Copilot CLI

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How the AMD Knowledge Base Grounds Ross Responses

Vectorized Knowledge Base (RAG)



This vectorized knowledge base is available via the cloud or hosted on a local server for air-gapped environments

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Access Knowledge Base: Real Questions, Grounded Answers

System Architect

How to configure NoC for optimal bandwidth?

→ UG1388, UG1504, PG313

Use traffic classes (Best Effort, Isochronous, Low Latency) with bandwidth allocation per route

Which AMD Versal™ device supports my AI inference workload?

→ DS957, UG1504

Versal AI Edge (VE2302, VE2802) with integrated AI Engines for INT8/INT4 inference. 400+ TOPS

RTL Developer

How to constrain async clock domain crossings?

→ UG903, UG1387, UG949

Use XPM_CDC macros. Apply ASYNC_REG attributes
set_false_path / set_max_delay constraints

RTL coding best practices for synthesis?

→ UG901, UG1387, UG949

Follow UG901 guidelines. Use RAM_STYLE, SHREG_EXTRACT,
MAX_FANOUT attributes. Avoid latches

Powered by RAG—500+ User Guides, Application Notes & Knowledge Base Articles
UG903 · UG904 · UG908 · UG949 · UG1387 · UG1388 · UG1504 · PG313 · DS957 · AM011 · UG901 · AR Articles · and more

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Access Knowledge Base: Real Questions, Grounded Answers

Timing Closure Specialist

How to fix setup violations on high-fanout nets?

→ UG949, UG1387, AR#53430

Use report_high_fanout_nets after synthesis
Apply MAX_FANOUT or manual replication

What phys_opt_design techniques help timing?

→ UG904, UG949, UG1387

Post-place and post-route phys_opt_design
Use -fanout_opt,
-placement_opt, -retime

Hardware Debug Specialist

PDI boot failure troubleshooting on AMD Versal devices?

→ UG1304, UG908, AR#76655

Check boot mode pins (SW1). Verify PDI in JTAG/flash/SD
Boot via HW Manager. Check PMC error codes

DDR calibration debug using DDRMC in the AMD Vivado™ tool?

→ UG908

DDRMC debug always enabled in Versal devices. HW Manager reads calibration status, R/W window margins

Powered by RAG—500+ User Guides, Application Notes & Knowledge Base Articles
UG903 · UG904 · UG908 · UG949 · UG1387 · UG1388 · UG1504 · PG313 · DS957 · AM011 · UG901 · AR Articles · and more

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Deep Dive: Knowledge Database

COVERAGE
What's Inside

- Cross-referenced Answers
  - Web documentation
  - PDF user & product guides
  - Knowledge base (answer records)
  - Blog articles & design tips
  - Embedded wiki (Linux®, u-boot)
  - Github reference designs
- Product & Tool Coverage
  - AMD Versal™ AI Edge / Premium / HBM
  - Zynq™ UltraScale+™ MPSoC & RFSoC
  - AMD 7 Series, Spartan™ 6, Virtex™ 6/5 FPGAs
  - Vivado™ IDE, Vitis™ software platform, PetaLinux, ISE™ design suite

SMART TAGGING
How It's Organized

- Source Classification
  - Web docs
  - Knowledge base
  - PDF docs
  - Blog
  - Wiki
  - GitHub
- Document ID Taxonomy
  - UG
  - PG
  - DS
  - WP
  - AR#
  - AM
- Section-Level Indexing
  - Deep links to exact doc sections
  - UG903 → Async Clock groups
  - UG1304 → PLM responsibilities
  - UG949 → CDC methodology

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Deep Dive: Knowledge Database

INTELLIGENT SEARCH
How It Finds Answers

- Dual Retrieval Engine
  - Keyword matching for Tcl commands, register names, error codes
  - Semantic understanding for natural language questions
- Flexible Search Modes
  - Precision: exact term lookup
  - Balanced: keyword + semantic
  - Exploration: intent-based discovery
- Result Control
  - Adjustable depth: focused answer to comprehensive survey
  - Every result cites doc ID + URL

CONTEXT AWARENESS
Why It's Better

- Cross-Referenced Answers
  - Results link to related docs
  - e.g., UG949 → UG903 → UG1387
  - Design advisories auto-surfaced
- Design-Phase Mapping
  - Matches query to design stage
  - RTL → Synth → Impl → Debug
  - Phase-appropriate doc priority
- Actionable Responses
  - Full context excerpts, not titles
  - Tcl commands included inline
  - Step-by-step guidance from UGs

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AMD Vivado MCP Tool Reference

☷ vivado_doc_search

RAG-powered documentation search—foundation for all agent responses

19 tools • 6 categories • MCP Protocol

Session

- vivado_start
  Launch Vivado™ tool (Tcl or GUI mode)
- vivado_stop
  Close the active session
- vivado_connect
  Connect to external webserver
- vivado_list_sessions
  List all active sessions
- vivado_cleanup
  Remove stale sessions & orphans

Monitoring

- vivado_status
  Process status & command queue
- vivado_log_messages
  Parse log for errors & warnings
- vivado_run_status
  Check synth/impl run progress
- vivado_run_health
  Detect & recover stuck runs

LSF / HPC

- vivado_lsf_start
  Launch Vivado IDE on cluster via bsub
- vivado_lsf_status
  Check job: PEND / RUN / DONE / EXIT
- vivado_lsf_kill
  Kill a job by ID or session ID

Execution

- vivado_execute
  Run any Tcl command in Vivado IDE
- vivado_history
  Search session command history

Utilities

- vivado_display_check
  Verify X11 display before GUI launch
- vivado_client_info
  Get connected MCP client details

Productivity

- vivado_todos
  Manage a persistent task/todo list
- vivado_show_todos
  Display todos in formatted sidebar view

Most tools are available today; a few are still in development

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Summary

01 The AMD Ross™ agentic AI assistant for Vivado™ Design Suite enables users to search AMD documentation, drive the Vivado tools, and execute complete FPGA workflows using plain English

02 Acceleration of debugging and root cause analysis, enabling complete design generation from a single prompt

03 Built on MCP and agent skills, making it IDE-agnostic and LLM-agnostic out of the box

04 All capabilities are unified within a single server, enabling documentation search, Tcl execution, session management, and session monitoring

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Agentic AI Workflows for Advanced Adaptive SoC Design

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Agenda

1 **Execution & Workflows**

- Vivado™ Execute: Closed-loop workflows (batch + GUI)
- Agent Skills: Repeatable, deterministic workflows

2 **Deployment & Ecosystem**

- Reduce OPEX: Local models, local knowledge base (KB), air-gapped solution
- How to get started + resources

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Agenda

1 **Execution & Workflows**

- Vivado™ Execute: Closed-loop workflows (batch + GUI)
- Agent Skills: Repeatable, deterministic workflows

2 **Deployment & Ecosystem**

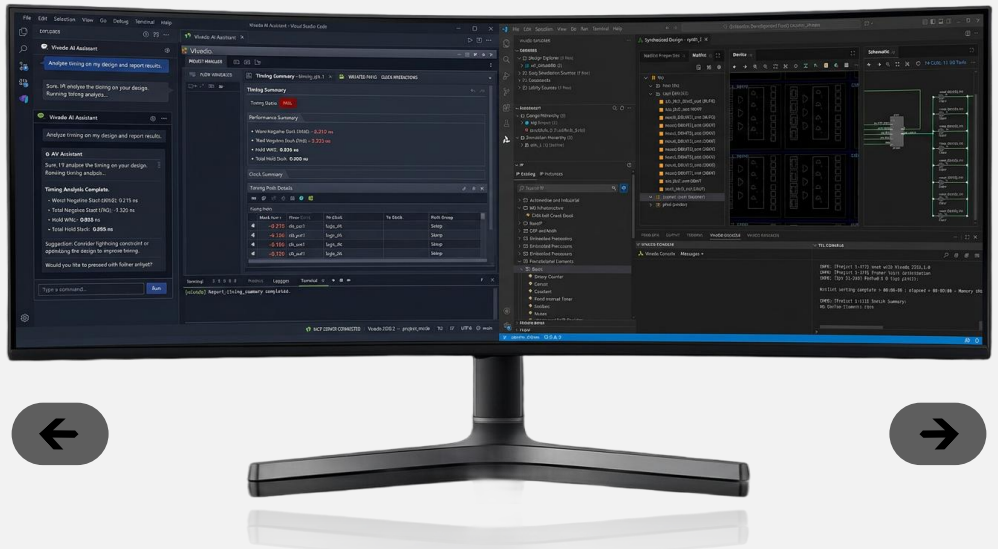
- Reduce OPEX: Local models, local knowledge base (KB), air-gapped solution
- How to get started + resources

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Working Alongside Developers

Connecting to live projects via CLI or IDE of choice

VS Code, Claude Code, Cursor, Devin, etc.



AMD Ross

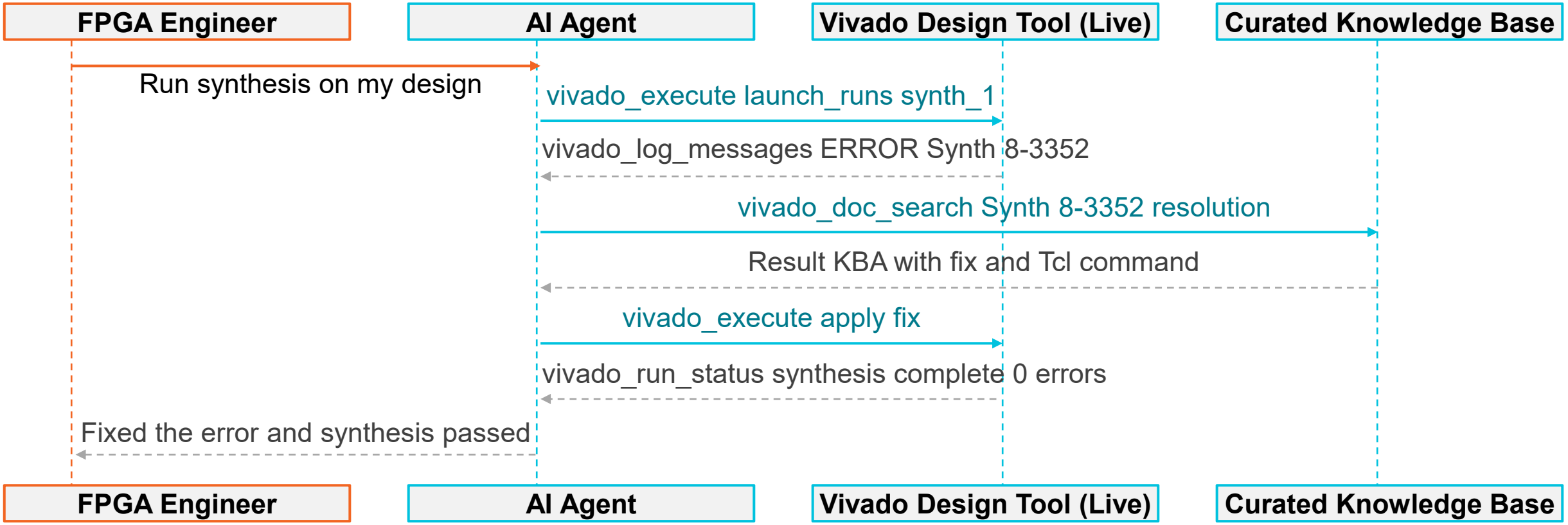
MCP Servers

- Works alongside the live project
- Execute workflows with full project access, without requiring Tcl scripting expertise
- Controls permissions and runs parallel sessions
- Operates in GUI for real-time interaction or batch-mode

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Closed-Loop Value: From Error to Resolution

What makes this transformative for customers is not the KB alone—it is the **tight integration** with live AMD Vivado™ design tool session control
A typical AI-assisted debug cycle is as follows:



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AMD Vivado MCP: Drives Vivado IDE in Batch or GUI Mode

Supports **same MCP tools**, **same AI agent**, and **choose the interface that fits your workflow**

BATCH MODE

- Headless
- CI/CD Pipelines
- Regression Flows



```
—— Vivado™ MCP—Tcl Batch Session ——
$ vivado_start session_type="batch"
  ✓ Vivado started in Tcl mode (PID 38012)

$ vivado_execute "open_project top_design.xpr"
  top_design

$ vivado_execute "launch_runs synth_1"
  ✓ synth_1 launched

$ vivado_run status="synth_1"
  synth_1: 100% Complete

$ vivado_execute "launch_runs impl_1"
  ✓ impl_1 launched

$ vivado_log_messages
  0 Errors, 2 Warnings, 14 Info

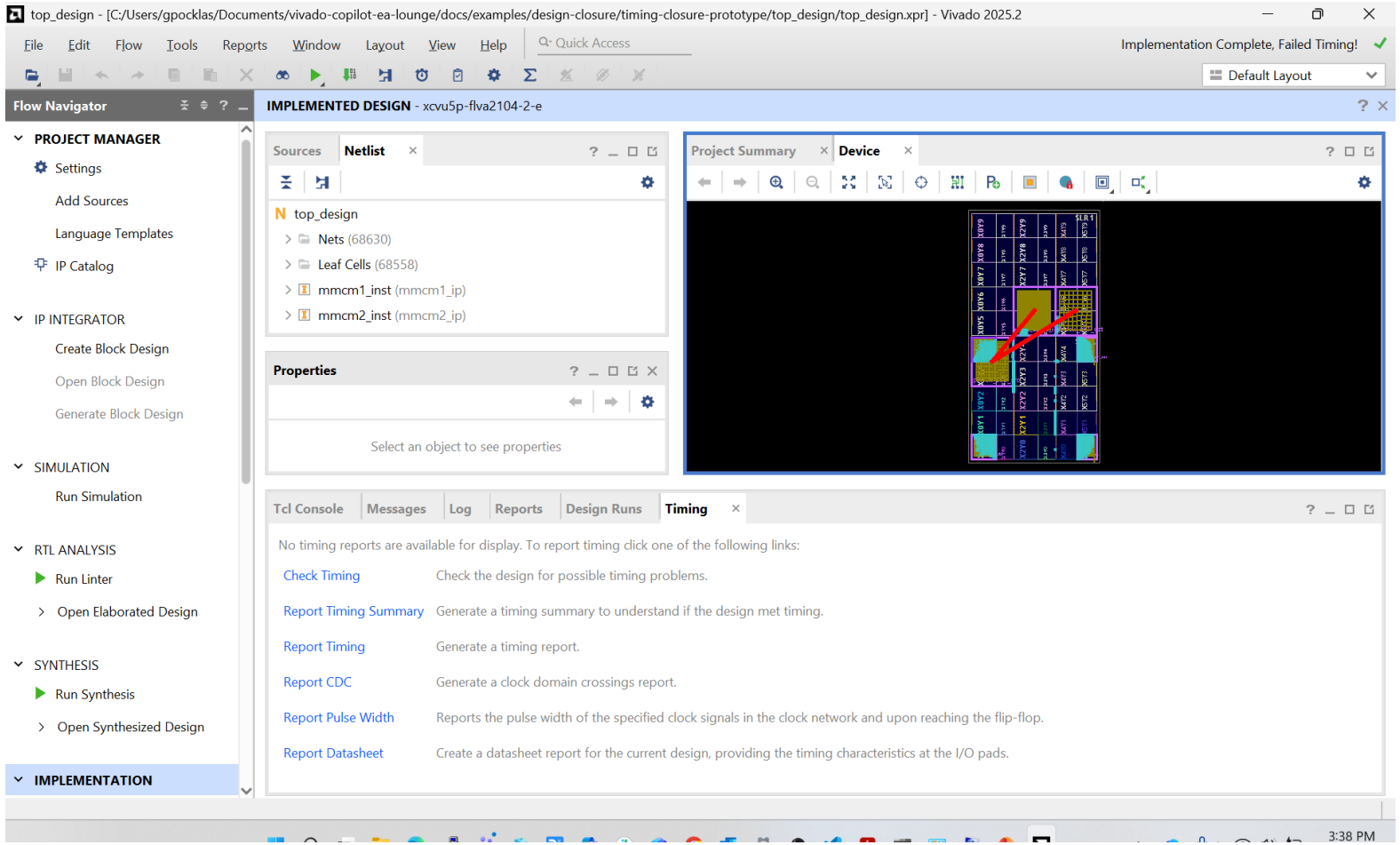
$ vivado_execute "report_timing_summary"
  WNS = 0.123ns—All timing constraints met
$
```


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AMD Vivado MCP: Drives Vivado IDE in Batch or GUI Mode

GUI MODE

Visual Debug • Interactive • Remote VNC



vivado_start gui_mode=true

→ Full Vivado IDE with device view, timing reports, schematic, and interactive debug—via VNC or local display

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Leveraging Agent Skills to Create Reliable Vivado IDE Workflow

.claude/skills/congestion-analysis/SKILL.md

```
---
name: congestion-analysis
version: 1.0.0
vivado: 2025.1+
duration: 3-5 minutes
difficulty: intermediate
execution: autonomous
---
# Congestion Analysis
## Mandatory Workflow

### Step 1: Generate Congestion Report
report_design_analysis ¥
  -congestion -file congestion.rpt
```

```
### Step 2: Identify Hotspot Regions
report_design_analysis ¥
  -congestion -min_congestion_level 5
# Parse North/South/East/West overflow
# Map hotspots to clock regions

### Step 3: Root Cause Analysis
report_utilization ¥
  -pblocks -file util_by_region.rpt
# Correlate LUT/FF/BRAM density
# with congested regions

### Step 4: Recommend Fixes
  → Pblock floorplanning
  → SLR assignment for SSI devices
  → Synthesis retiming / restructuring
  → Phys-opt fanout replication
## Output: REPORT.md with fix commands
```


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Leveraging Agent Skills to Create Reliable Vivado IDE Workflow

Autonomous	Runs with zero user intervention: the agent executes all five steps and writes the report automatically
Structured Workflow	Real Vivado™ Tcl commands in every step—no hallucinated syntax; tested against live designs
Self-validating	Design-specific fixes with real cell names, clock names, and constraints—no generic placeholders
Chainable	Skills declare upstream/downstream dependencies: Run congestion → then fanout-opt automatically

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Key Components of Ross: Why Skills Help

Skills provide LLMs with subject matter expertise and specific AMD guidelines and use specific context they can load on demand

Without Skills, LLMs…

- Need more prompting
- Require attached reference files
- Produce more inconsistent results
- Consume more tokens
- Access out-of-date documentation

Use Cases

Domain Expertise	• AMD tools methodology & best practices • Years of expert knowledge in executable form
Repeatable Workflows	• Consistent, predictable results • Proven step-by-step methodology • Reduced trial-and-error guessing
Efficient Token Usage	• Support for custom design-pattern skills • Reusable skills across teams and organizations • Agent extensibility without code changes

Skills Bridge the Methodology Gap

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Key Components of Ross: How Skills Work…

File System

Skills stored as markdown files

SKILL with progressive disclosure of instructions & resources, e.g., scripts

Metadata: Always loaded

```
---
name: csim
description: Run C Simulation to verify kernel correctness and collect loop trip count profiles for synthesis.
allowed-tools: Bash C Simulation , Read
---
```

Instructions: Loaded when triggered

```
# Skill: csim

Run C Simulation (csim) to verify the kernel and collect loop trip count profiles.

## When to use

Run csim:
- After modifying `src/*.cpp` or `src/*.h`, to confirm the kernel still produces correct results.
- Before csynth, to generate loop trip count data that improves HLS latency/throughput estimates.
- When debugging: a csim failure (non-zero exit or "FAIL" output) means the kernel logic is broken.

## How to run

```bash
vitis-run --mode hls --config $CONFIG --work_dir $TOP_FUNCTION --csim
```

This runs `vitis-run --mode hls --config $CONFIG --work_dir $TOP_FUNCTION --csim`, which:
```

LLM

Executes expert workflows using skill **instructions**

- Generating code
- Running tools
- Validating results

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Key Components of Ross: How Skills Work…

File System

Skills stored as markdown files

SKILL with progressive disclosure of instructions & resources, e.g., scripts

Resources: Loaded as needed

```
## Scripts

Three helper scripts live in `.claude/skills/synth/scripts/`. Run them from the repo root.

### `report_summary.py` — compact synthesis report

```bash
python3 .claude/skills/synth/scripts/report_summary.py [report.rpt]
```

LLM

Executes expert workflows using skill **instructions**

- Generating code
- Running tools
- Validating results

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```

AMD Vivado IDE Agent Skills—Continued Support

Design Analysis

- **synth-design-analysis**—Post-synthesis QoR & resource assessment
- **opt-design-analysis**—Logic optimization effectiveness review
- **phys-opt-design-analysis**—Physical optimization analysis
- **route-design-analysis**—Routing congestion & timing closure
- **congestion-analysis**—Routing congestion hotspot analysis
- **fanout-opt**—High-fanout net optimization & replication
- **reset-architecture-review**—Reset tree & async reset verification
- **dfx-floorplan-review**—DFX partition floorplan validation
- **custom-drc-ep**—Custom design rule checks

Clock & Timing

- **clock-domain-crossing-review**—CDC path analysis & XPM_CDC recommendations
- **clock-tree-topology-review**—Clock architecture & MMCM/PLL topology
- **clock-skew-uncertainty-optimization**—Skew & uncertainty budget optimization
- **clock-placer-failure-debug**—Debug clock placement failures
- **clock-region-placer-failure-ep**—Clock region resource exhaustion

Hardware Debug

- **bd-ila-insertion**—Block design ILA probe insertion
- **noc-compiler-failure-analysis**—NoC compiler failure root-cause debug
- **noc-performance-monitoring**—NoC traffic analysis & bandwidth tuning
- **pcie-link-analysis**—PCIe® link training & LTSSM debug
- **ddr-calibration-debug**—DDR4/5 calibration failure analysis

Platform Specific

- **versal-timing-closure-methodology** — Versal™ adaptive SoC timing closure best practices
- **versal-multi-run-analysis** — Multi-strategy run comparison & selection

AMD is developing high-value agent skills to strengthen the Vivado™ MCP value proposition

Work in progress—some of these skills are available now and more skills will be available throughout the year | Users are free to create custom skills tailored to their own designs

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Agenda

1 Execution & Workflows

- Vivado™ Execute: Closed-loop workflows (batch + GUI)
- Agent Skills: Repeatable, deterministic workflows

2 Deployment & Ecosystem

- Reduce OPEX: Local models, local knowledge base (KB), air-gapped solution
- How to get started + resources

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Air-gapped, Local Deployment Options

For Security-sensitive or Cost-conscious Team Environments

- Run AMD Ross™ knowledge base and MCPs locally without relying on the cloud
- Keep prompts, project context, and design data within controlled infrastructure

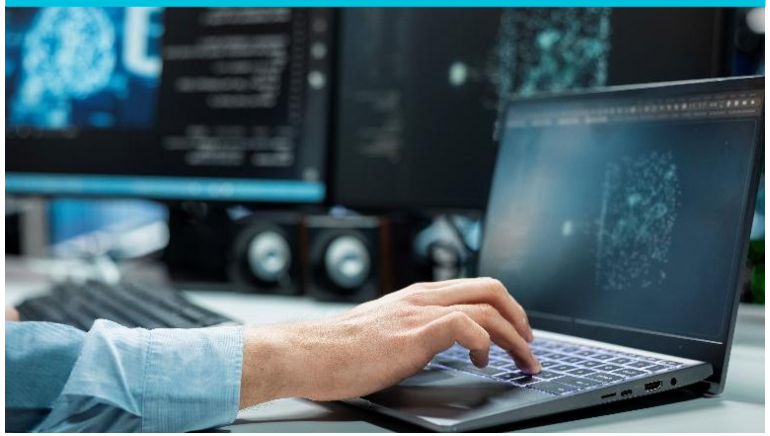
IT Security Policies

Keep sensitive data within the enterprise



Control Inference Costs

Avoid recurring cloud API and token fees



Restricted Environments

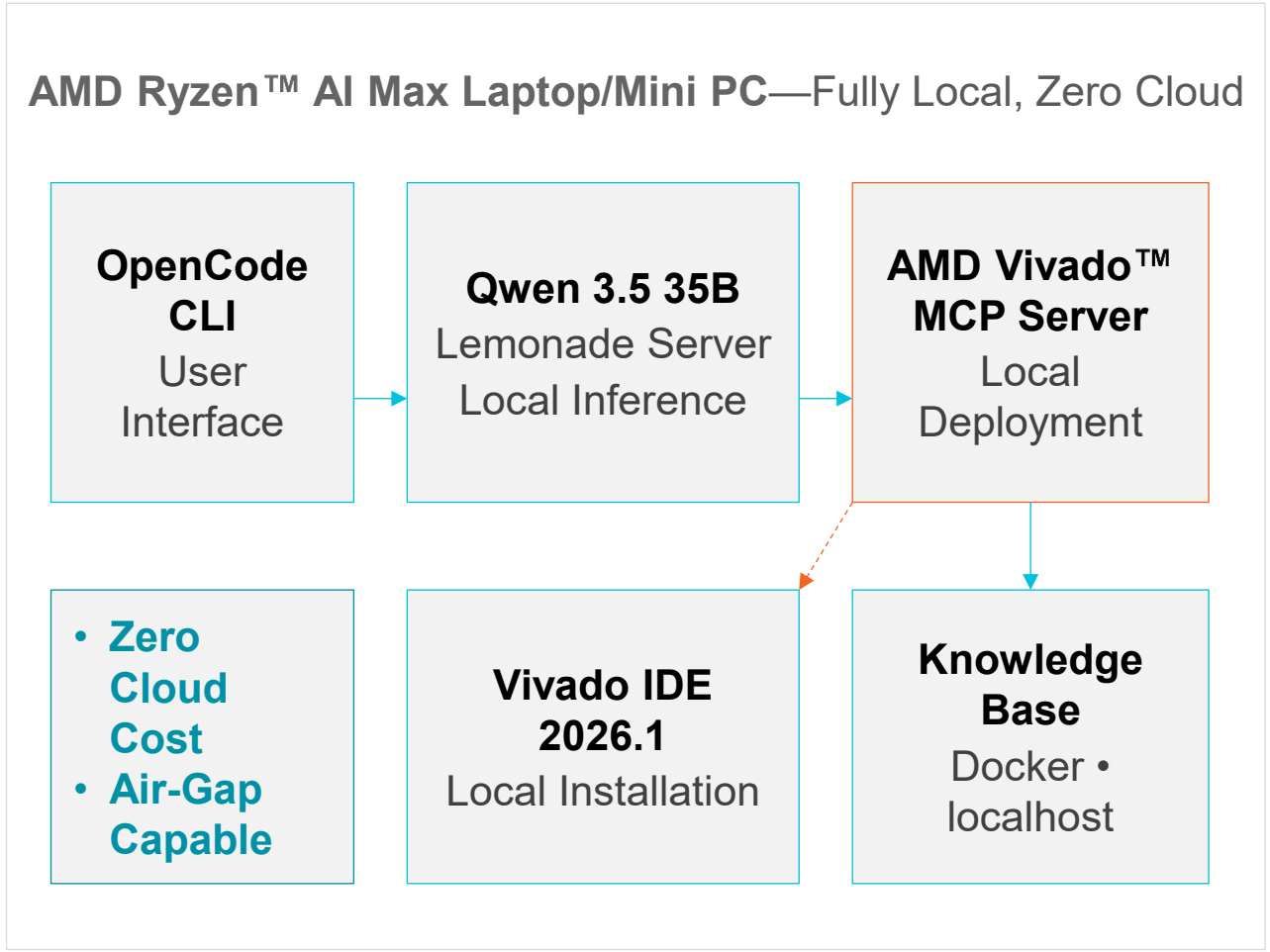
Work where cloud access is limited



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Recognizing the Importance of Reducing OPEX



Full Local Inference

Qwen 3.5 35B on AMD silicon via **Lemonade Server**[1]—no API keys, no subscriptions

Same MCP Tools

The Vivado MCP toolset interface remains consistent across local and cloud deployments

Modular Stack

Each component independently deployable—swap LLM, upgrade KB, or scale as needed

Secure by Design

All data stays on-premises—ideal for defense, government, and IP-sensitive environments

1. Lemonade is a **local AI server** that runs models on your own hardware and exposes an **OpenAI-compatible API**

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Setting Expectations: Local LLM vs. Frontier Models

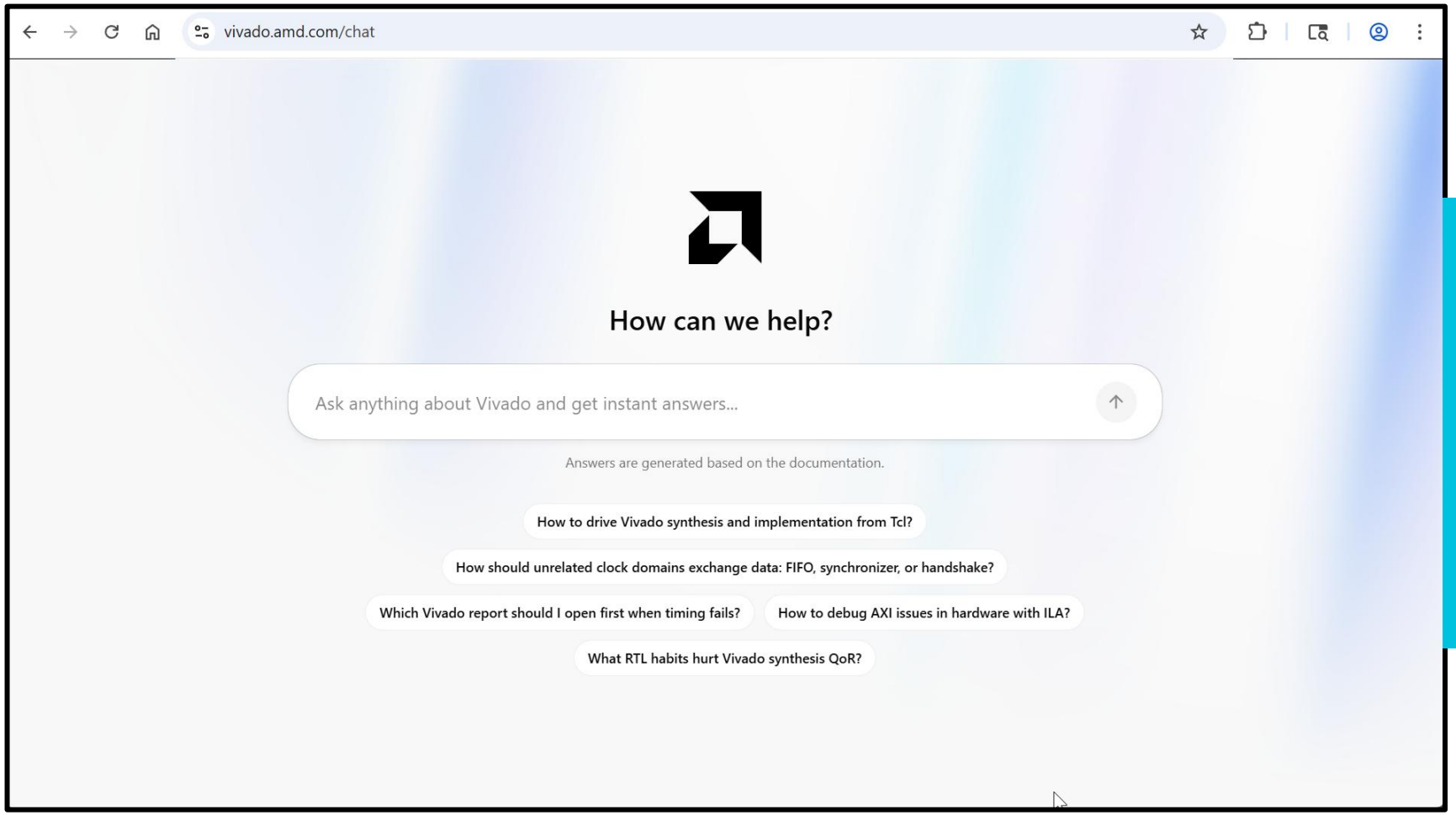
Dimension	Local LLM (On-Prem)	Frontier Model (Cloud)
Strength	Great for documentation Q&A, tool execution, repeatable workflows	Best for deep reasoning, longer code generation, fewer iterations
Tunning for Vivado™ MCP Parameters	Needed for context window management, token consumption, and prompt sizing	Not needed
Cost	Fixed infrastructure cost; no per-token/API fees	Usage-based (tokens, requests, subscriptions)
Data & Privacy	Prompts, designs, and logs stay on-prem	Data leaves the network (subject to provider policies)
Deployment	Can run offline/air-gapped	Requires network/service access

- MCP architecture is LLM-agnostic
- Allowing users to use local or cloud models while retaining the same knowledge base, MCP tools, and workflows

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A One-Stop Shop for AMD Vivado IDE Queries

Access the AMD knowledge base via a web portal: vivado.amd.com/chat



- Connects to the same AMD network-hosted documentation database
- Delivers retrieval-augmented generation (RAG) powered, citation-backed outputs

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Let’s Get Started . . . Steps

1 Prerequisites

- Download, install, and license the AMD tools
- Install an MCP-compatible IDE/CLI client
- Ensure access to your preferred LLM

2 Download the MCP Server

- Sign in at <amd.com> with your AMD credentials
- Download the MCP server build for your OS
- Configure your IDE/CLI client to connect to MCP

3 Get Skills and Examples

- Clone agent skills and example designs from github.com
- Install the skills in the appropriate location for your setup
- Confirm the skills load in your client

4 Start Your AI Journey

- Follow the design examples end to end
- Iterate with AMD tools in your daily loop

End state: A working MCP setup with AMD skills installed and your first agentic design running

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Summary

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AMD Ross™ agentic AI assistant for Vivado™ tools boosts design productivity through:

- Automated, agent-driven workflows for error detection, fixing, and validation
- Scalable, repeatable FPGA design via agent skills
- Flexible deployment (local, cloud, air-gapped)
- Seamless integration with Vivado tools
- Real-time, context-aware insights for faster debugging and optimization

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